

## Lecture 7: Black-Scholes and Delta Hedging

**Call/Put Options:** Let  $K$  be the strike price.

Assume current time is 0 and expiration is at time  $t$ .

- Call option payoff at expiration:

$$C(t) = \max(S(t) - K, 0)$$

如果股票价格  $S(t)$  高于执行价  $K$ ，就有正收益；否则一文不值。

- Put option payoff at expiration:

$$P(t) = \max(-S(t) + K, 0)$$

## Black-Scholes Pricing Formula

Black-Scholes 定理的核心思想是：在没有套利的前提下，一个欧式期权的价格可以表示为其到期价值的 风险中性期望值。

Let  $\mathbb{E}$  denote expectation under the risk-neutral measure. Then:

$$C(0) := \mathbb{E}[C(t)], \quad P(0) := \mathbb{E}[P(t)]$$

These represent the present values of the expected payoffs under the risk-neutral measure.

If we assume:

- No risk-free interest rate (i.e.,  $r = 0$ ),
- No drift in the stock price (i.e., we model under risk-neutral dynamics),

then the asset price evolves as:

$$S(t) = S(0) \cdot \exp\left(-\frac{1}{2}\sigma^2 t + \sigma\sqrt{t}Z\right), \quad Z \sim \mathcal{N}(0, 1)$$

This corresponds to the log-normal model of geometric Brownian motion under risk-neutral measure when  $r = 0$ .

## Black-Scholes Closed-Form Solution (Zero Interest Rate)

Using the assumptions above, the option prices at time 0 are:

$$\begin{aligned} C(0) &= S(0) \cdot \Phi(d_1) - K \cdot \Phi(d_2) \\ P(0) &= -S(0) \cdot \Phi(-d_1) + K \cdot \Phi(-d_2) \end{aligned}$$

Where:

- $\Phi(\cdot)$  is the cumulative distribution function (c.d.f.) of the standard normal distribution  $\mathcal{N}(0, 1)$ .

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$$d_1 = \frac{\ln\left(\frac{S(0)}{K}\right) + \frac{1}{2}\sigma^2 t}{\sigma\sqrt{t}}, \quad d_2 = d_1 - \sigma\sqrt{t}$$

## Remarks

- These formulas are derived under the assumption that the market is frictionless and that options are priced using the no-arbitrage principle under a risk-neutral measure.
- Frictionless market: 买卖资产不需要支付交易成本（如佣金、税收等），可以无限小时时间间隔内交易；
- No-arbitrage: 不存在无风险套利机会；
- When  $r \neq 0$ , the full Black-Scholes formula includes the discounting factor  $e^{-rt}$  in front of the strike price term.
- Delta hedging is a technique derived from this model, used to hedge against small movements in the price of the underlying asset by holding a position proportional to the derivative (delta) of the option price with respect to the asset.
- Delta Hedging 是基于 Black-Scholes 模型推导出的风险管理手段；它的目标是对冲股票价格的小幅波动风险；方法是持有一部分股票，使得整个投资组合对股票价格的导数（Delta）为零，达到“Delta 中性”状态。

## Black-Scholes Model with Drift and Risk-Free Interest Rate

### Black-Scholes Assumptions

Black-Scholes 模型成立的五个核心假设. The Black-Scholes model assumes:

- The stock price follows a geometric Brownian motion. 股票价格服从几何布朗运动（GBM）数学形式：

$$dS = \mu S dt + \sigma S dW$$

解释：

- 股票收益率是连续正态分布；
- 这是金融建模中的标准假设；
- 导致股票价格呈 **log-normal** 分布。
- Markets are frictionless, and there is continuous trading. 市场无摩擦，允许连续交易
  - 没有税费和交易成本；
  - 投资者可以无限频率地调仓（重要前提，用于推导 Delta Hedging）。
- The risk-free interest rate  $r$  is constant.
  - 银行等无风险资产给的利率是已知的常数；
  - 所以可以用  $e^{-rt}$  折现未来现金流。
- No arbitrage opportunities exist. 不存在套利机会
  - 没有可以无风险赚取利润的组合；
  - 这是整个 Black-Scholes 模型（和风险中性定价）的数学基础。

- The asset pays no dividends. 股票不支付股息
  - 股票在持有期间不会带来额外现金流;
  - 如果有分红, 则 Black-Scholes 公式要做调整 (加一个 dividend yield) 。

## 基本假设: 股票价格模型 Stock Price Dynamics

Under the **real-world measure** (with drift  $\mu$ ), the stock price evolves as:

$$S(t) = S(0) \cdot \exp \left[ \left( \mu - \frac{1}{2}\sigma^2 \right) t + \sigma\sqrt{t}Z \right], \quad Z \sim \mathcal{N}(0, 1)$$

This models the actual evolution of the stock, where:

- $\mu$  is the expected return (drift), 股票的预期收益率 (drift)
- $\sigma$  is the volatility.
- 实际上是几何布朗运动 (Geometric Brownian Motion) 的解

## Discounted Stock Price under Risk-Neutral Measure

To compute fair option prices, we move to the **risk-neutral measure**, where the expected return  $\mu$  is replaced by the risk-free rate  $r$ .

Then, the **discounted price** at  $t = 0$  is:

$$e^{-rt}S(t) = S(0) \cdot \exp \left[ \left( \mu - r - \frac{1}{2}\sigma^2 \right) t + \sigma\sqrt{t}Z \right]$$

- 如果我们把股票价格  $S(t)$  贴现到今天  $t = 0$ ,
- 那么它的增长率是  $\mu - r$ , 加上波动项;
- 如果我们想让  $e^{-rt}S(t)$  成为一个 **Martingale** (公平游戏), 那么我们就必须让  $\mu = r$ 。

Under the risk-neutral measure, the stock follows:

$$S(t) = S(0) \cdot \exp \left[ \left( r - \frac{1}{2}\sigma^2 \right) t + \sigma\sqrt{t}Z \right]$$

这是 **Black-Scholes** 模型中的核心假设:

- 股票在风险中性世界下的对数收益是正态分布;
- 分布中的 drift 从  $\mu$  换成了  $r$ , 使得我们能计算期权的无套利价格。

## Call Option Pricing Formula

Suppose a European call option expires at time  $t$ . Its payoff is:

$$\max(S(t) - K, 0)$$

To compute the price of the call at time 0, we discount the expected payoff under the risk-neutral measure:

$$C(0) = e^{-rt} \mathbb{E}[\max(S(t) - K, 0)]$$

- 这个公式是 Black-Scholes 理论的根本出发点;
- 先在风险中性世界下取期望, 再用  $e^{-rt}$  贴现回来;
- 它不需要知道投资者的风险偏好;
- 所有定价完全取决于  $S(0)$ ,  $K$ ,  $r$ ,  $t$ ,  $\sigma$  和 Black-Scholes 模型结构。

This represents the expected value of the option's payoff, discounted back from expiration  $t$  to time 0 using the risk-free rate  $r$ .

## Closed-form Black-Scholes Formula

Under the assumptions above, the closed-form solution for the European call and put options are:

$$\begin{aligned} C(0) &= S(0)\Phi(d_1) - Ke^{-rt}\Phi(d_2) \\ P(0) &= Ke^{-rt}\Phi(-d_2) - S(0)\Phi(-d_1) \end{aligned}$$

Where:

$$d_1 = \frac{\ln(S(0)/K) + (r + \frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}}, \quad d_2 = d_1 - \sigma\sqrt{t}$$

And  $\Phi(\cdot)$  is the cumulative distribution function (c.d.f.) of the standard normal distribution  $\mathcal{N}(0, 1)$ .

- $\Phi(d_1)$ : 在风险中性世界中, 期权到期时股票价格高于执行价的**概率加权**;
- $\Phi(d_2)$ : 实际行权时被折现的执行价的**风险中性概率**;
- 所以整个公式是:

当前股票价值  $\times$  盈利可能性 折现执行价  $\times$  被行权概率

## Overview of Risk-Neutral Pricing

基本假设: 可以连续地购买/卖出股票 (包括小数股), 从而构造出一个复制组合 (replicating portfolio)。

**Assumption:** It is possible to continuously buy/sell the underlying stock in fractional shares, and this can be done in a way that creates a replicating portfolio. That is, for each  $T \in [0, t]$ , we can construct a portfolio  $X(T)$  such that:

$$X(T) = C(T) = \mathbb{E}_{\mathbb{Q}} \left[ e^{-r(t-T)} \max(S(t) - K, 0) \mid \mathcal{F}_T \right]$$

- $X(T)$ : 我们通过动态持仓构造出来的投资组合的价值;
- 它在任意  $T \in [0, t]$  时都等于期权的价值;
- 这个组合的存在使得我们能使用**风险中性期望**来定价期权。
- This equation represents the price of a European call option at time  $T$ .
- The effect of the replication strategy is to eliminate the drift term  $\mu$ .
- $X(0)$  (the value of the portfolio at time 0) must equal the value of the call option at time 0.

Thus, under the risk-neutral measure  $\mathbb{Q}$ , the stock is modeled to grow at the risk-free rate  $r$  instead of the drift  $\mu$ .

## Stock Price Dynamics under Risk-Neutral Measure

Under the risk-neutral measure, the stock price follows:

$$S(t) = S(0) \cdot e^{(r - \frac{1}{2}\sigma^2)t + \sigma\sqrt{t}Z}, \quad Z \sim \mathcal{N}(0, 1) \quad (1)$$

## Pricing the Call Option

We want to compute the time-0 value of a European call option expiring at time  $t$  with strike  $K$ . The payoff is:

$$\max(S(t) - K, 0)$$

The present value (at  $t = 0$ ) of the call is:

$$C(0) = e^{-rt} \mathbb{E}_{\mathbb{Q}} [\max(S(t) - K, 0)]$$

## Deriving the Black-Scholes Formula

Using the log-normal distribution of  $S(t)$ , we derive the formula:

$$C(0) = e^{-rt} (\mathbb{E}[S(t)] \cdot \Phi(d_1) - K \cdot \Phi(d_2))$$

By algebraic manipulation:

$$C(0) = e^{-rt} (S(0) \cdot e^{rt} \Phi(d_1) - K \Phi(d_2))$$

$$\Rightarrow C(0) = S(0) \Phi(d_1) - K e^{-rt} \Phi(d_2)$$

## Black-Scholes Put Formula and Parity

The price of a European put option is:

$$P(0) = Ke^{-rt}\Phi(-d_2) - S(0)\Phi(-d_1)$$

The **Call-Put Parity** relation is:

$$C(t) - P(t) = S(t) - K \Rightarrow C(0) - P(0) = S(0) - Ke^{-rt}$$

## Definitions of $d_1$ and $d_2$

$$d_1 = \frac{\ln\left(\frac{S(0)}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)t}{\sigma\sqrt{t}}, \quad d_2 = d_1 - \sigma\sqrt{t}$$

## Conclusion

We aim to compute:

$$\mathbb{E}[\max(S(t) - K, 0)]$$

discounted to time 0 using  $e^{-rt}$ . This results in the celebrated Black-Scholes formula.

## Delta Hedging

Delta 对冲 是一种保护利润的方法，特别适用于卖出（short）Call 或 Put 期权的情况。通过动态调整股票仓位，使得整个投资组合对股票价格的小幅变化不敏感（即 delta 中性）。

**Overview:** Delta hedging is a practical method for protecting profits, especially when selling call or put options. It is implemented as a discrete process, but its theoretical foundation lies in continuous-time hedging — a key idea in Black-Scholes option pricing.

## Objective:

Determine the sensitivity of the call option price with respect to the underlying asset price, i.e.,

$$\frac{\partial C(0)}{\partial S(0)}$$

## Call Option Formula:

$$C(0) = S(0)\Phi(d_1) - Ke^{-rt}\Phi(d_2)$$

Let  $\varphi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$  be the standard normal probability density function (pdf), and

$$\Phi(y) = \int_{-\infty}^y \varphi(z) dz$$

**Differentiating  $C(0)$ :**

We compute:

$$\frac{\partial C(0)}{\partial S(0)} = \Phi(d_1) + S(0) \cdot \frac{\partial \Phi(d_1)}{\partial S(0)} - Ke^{-rt} \cdot \frac{\partial \Phi(d_2)}{\partial S(0)}$$

Using the chain rule:

$$\frac{\partial \Phi(d)}{\partial S(0)} = \varphi(d) \cdot \frac{\partial d}{\partial S(0)}$$

So we get:

$$\frac{\partial C(0)}{\partial S(0)} = \Phi(d_1) + S(0) \cdot \varphi(d_1) \cdot \frac{\partial d_1}{\partial S(0)} - Ke^{-rt} \cdot \varphi(d_2) \cdot \frac{\partial d_2}{\partial S(0)}$$

Recall:

$$d_2 = d_1 - \sigma\sqrt{t} \Rightarrow \frac{\partial d_2}{\partial S(0)} = \frac{\partial d_1}{\partial S(0)}$$

Thus:

$$\frac{\partial C(0)}{\partial S(0)} = \Phi(d_1) + \frac{\partial d_1}{\partial S(0)} [S(0)\varphi(d_1) - Ke^{-rt}\varphi(d_2)]$$

**Algebraic simplification:** It turns out:

$$S(0)\varphi(d_1) = Ke^{-rt}\varphi(d_2) \Rightarrow \frac{\partial C(0)}{\partial S(0)} = \Phi(d_1)$$

**Put Option Delta via Put-Call Parity**

By put-call parity:

$$C(0) - P(0) = S(0) - Ke^{-rt} \Rightarrow \frac{\partial C(0)}{\partial S(0)} - \frac{\partial P(0)}{\partial S(0)} = 1$$

So:

$$\frac{\partial P(0)}{\partial S(0)} = \Phi(d_1) - 1$$

**Delta-Hedged Portfolio Example**

如何用 Delta-Hedging 构造一个对冲组合 (Delta-Hedged Portfolio) , 并实现对小幅价格变化不敏感 (delta 中性) 。

Suppose we sell 100 call options at time 0 (expiring at  $t$ ), receiving total premium  $P$ . Then:

**卖出 100 份欧式看涨期权**

- 时间:  $t = 0$
- 到期时间:  $t$
- 卖出价格为  $P$  (premium) , 即总共收到  $100 \times P$
- 每份期权当前价值为  $C(0)$

初始投资组合价值  $X(0)$ . **Portfolio value at time 0:**

$$X(0) = 100(P - C(0))$$

这个组合由两个部分组成:

1. 收到的总 premium:  $100 \cdot P$
2. 未来需要履约的义务 (期权被行权时损失):  $-100 \cdot C(0)$

对股票价格  $S(0)$  的敏感度 (也就是组合的 Delta). **Sensitivity with respect to  $S(0)$ :**

$$\frac{\partial X(0)}{\partial S(0)} = -100 \cdot \frac{\partial C(0)}{\partial S(0)} = -100\Phi(d_1)$$

因为 Call 的 Delta 是:

$$\Delta_{\text{Call}} = \frac{\partial C(0)}{\partial S(0)} = \Phi(d_1)$$

**Interpretation:** The portfolio is negatively correlated with  $S(0)$ .

**To delta hedge:** buy  $100 \cdot \Phi(d_1)$  shares of the underlying asset.

组合价值随股票上涨而下降, 因为你卖出的是看涨期权 (股票涨你就可能亏钱)。(负相关结构)

**Hedged portfolio value:**

$$X_{\text{hedge}}(0) = 100(P - C(0)) + 100 \cdot S(0) \cdot \frac{\partial C(0)}{\partial S(0)}$$

**So:**

$$\frac{\partial X_{\text{hedge}}(0)}{\partial S(0)} = 0$$

This means the hedged portfolio is insensitive to small changes in  $S(0)$  — i.e., delta-neutral.

**如何 Delta 对冲?**

**目标:** 让整个组合的对  $S(0)$  的导数 (Delta) 为 0, 即变成 “Delta-neutral”。

✓ **策略:** 买入  $100 \cdot \Phi(d_1)$  股的股票。

**Delta-hedged 投资组合的价值**

现在组合变成:

- 收到的 premium:  $100(P - C(0))$
- 持有股票  $100 \cdot \Phi(d_1)$  股, 对应初始价值为  $100 \cdot S(0) \cdot \Phi(d_1)$

所以总资产变成:

$$X_{\text{hedge}}(0) = 100(P - C(0)) + 100 \cdot S(0) \cdot \Phi(d_1)$$

**对冲成功的验证** 我们计算这个组合的 Delta:

$$\frac{\partial X_{\text{hedge}}(0)}{\partial S(0)} = -100 \cdot \Phi(d_1) + 100 \cdot \Phi(d_1) = 0$$

```
[9]: ## Function experiment
S0 = 100      # 当前股价 $100
K = 110      # 执行价 $110
sigma = .3    # 年化波动率 30%
tte = 2       # 两年到期
rate = .03    # 无风险利率 3%

bs_call(S0, K, sigma, t = tte, r = rate), bs_put(S0, K, sigma, t = tte, r = rate)

[9]: (15.362749102373328, 18.956847796640695)
```

### 时间价值 vs 内在价值

- Call 的内在价值:  $\max(S_0 - K, 0) = \max(100 - 110, 0) = 0$
- Put 的内在价值:  $\max(K - S_0, 0) = 10$

可以看到, Put 的价格  $\approx 18.96$  包含了:

- \$10 的内在价值;
- \$8.96 的时间价值 (波动性提供的机会)

而 Call 的 \$15.36 完全来自时间价值。

### 波动率对价格的影响很大

因为  $\sigma = 0.3$  很高, 两个期权的价格都不低。这体现了:

**波动性越高, 期权越值钱** (尤其是 Out of the Money 期权)。

| 项目              | 数值      | 含义                       |
|-----------------|---------|--------------------------|
| Call 价格         | \$15.36 | 没有内在价值, 全部是时间价值          |
| Put 价格          | \$18.96 | \$10 是内在价值, \$8.96 是时间价值 |
| Put > Call      | 满足      | 因为当前股价低于执行价              |
| Put-Call Parity | 满足      | 数值完全匹配理论公式               |